

APPENDIX I

This part provides the analytic form of A_0, A_1, B_1, C_1 in the following formula:

$$\Delta\omega_{r,i}(t) = A_0 + A_1 e^{-\frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}}t} + B_1 e^{-\zeta\omega_n t} \cos(\omega_d t) + C_1 e^{-\zeta\omega_n t} \sin(\omega_d t) \quad (1)$$

When the sudden increase in the load: $\Delta P_L/s$ is added, the following $\Delta\omega_{r,i}(s)$ can be derived:

$$\Delta\omega_{r,i}(s) = -\frac{\Delta P_L K_{inertia,i} R T_R \omega_n^2}{2H_{w,i} \omega_{r,i,0} (D_{eq} R + 1)} \cdot \frac{(s + \frac{K_{droop,i}}{K_{inertia,i}})(s + \frac{1}{T_R})}{s(s + \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}})(s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad (2)$$

Denote $-\frac{\Delta P_L K_{inertia,i} R T_R \omega_n^2}{2H_{w,i} \omega_{r,i,0} (D_{eq} R + 1)} := M$. Then (2) can be decomposed into the following partial fractions:

$$\Delta\omega_{r,i}(s) = \frac{A_0}{s} + \frac{A_1}{s + \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}}} + \frac{B_1(s + \zeta\omega_n) + C_1\omega_n\sqrt{1-\zeta^2}}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (3)$$

(3) is just the Laplacian transform of (1).

A_0 can be obtained as:

$$A_0 = M \cdot \frac{2H_{w,i} K_{droop,i}}{3T_R K_{inertia,i} k_{MPPT} \omega_{r,i,0} \omega_n^2} \quad (4)$$

A_1 can be obtained as:

$$A_1 = M \cdot \frac{(s + \frac{K_{droop,i}}{K_{inertia,i}})(s + \frac{1}{T_R})}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \Big|_{s = -\frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}}} = M \cdot \frac{(-\frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}} + \frac{K_{droop,i}}{K_{inertia,i}})(-\frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}} + \frac{1}{T_R})}{-\frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}}((\frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}})^2 - \zeta\omega_n \frac{3k_{MPPT}\omega_{r,i,0}}{H_{w,i}} + \omega_n^2)} \quad (5)$$

In order to obtain B_1, C_1 , the following equations can be obtained:

$$\begin{aligned} M \cdot \frac{(-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2} + \frac{K_{droop,i}}{K_{inertia,i}})(-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2} + \frac{1}{T_R})}{(-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2})(-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2} + \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}})j2\omega_n\sqrt{1-\zeta^2}} \\ = \frac{B_1(-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2} + \zeta\omega_n) + C_1\omega_n\sqrt{1-\zeta^2}}{j2\omega_n\sqrt{1-\zeta^2}} \end{aligned} \quad (6)$$

Then it can be transformed into the following form:

$$\begin{aligned} B_1 j\omega_n\sqrt{1-\zeta^2} + C_1\omega_n\sqrt{1-\zeta^2} = \\ M \cdot \frac{(-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2} + \frac{K_{droop,i}}{K_{inertia,i}})(-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2} + \frac{1}{T_R})}{(-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2} + \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}})(-\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2})} \end{aligned} \quad (7)$$

Denote $s_1 = -\zeta\omega_n + j\omega_n\sqrt{1-\zeta^2}$. The following can be derived:

$$\begin{aligned}
& B_1 j \omega_n \sqrt{1-\zeta^2} + C_1 \omega_n \sqrt{1-\zeta^2} = \\
& M \cdot \frac{(-\zeta \omega_n + j \omega_n \sqrt{1-\zeta^2} + \frac{K_{droop,i}}{K_{inertia,i}})(-\zeta \omega_n + j \omega_n \sqrt{1-\zeta^2} + \frac{1}{T_R})}{(-\zeta \omega_n + j \omega_n \sqrt{1-\zeta^2} + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}})(-\zeta \omega_n + j \omega_n \sqrt{1-\zeta^2})} = M \cdot \frac{(s_1 + \frac{K_{droop,i}}{K_{inertia,i}})(s_1 + \frac{1}{T_R})}{(s_1 + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}})s_1} \\
& = M \cdot \frac{s_1^2 + (\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R})s_1 + \frac{K_{droop,i}}{T_R K_{inertia,i}}}{s_1^2 + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}s_1} = M \cdot \frac{s_1^2 + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}s_1 + (\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}})s_1 + \frac{K_{droop,i}}{T_R K_{inertia,i}}}{s_1^2 + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}s_1} \\
& = M \cdot (1 + \frac{(\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}})s_1 + \frac{K_{droop,i}}{T_R K_{inertia,i}}}{s_1^2 + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}s_1}) \\
& = M \cdot (1 + \frac{\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}}{s_1 + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}} + \frac{K_{droop,i}}{T_R K_{inertia,i}} \cdot \frac{1}{s_1^2 + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}s_1}) \\
& = M \cdot (1 + \frac{\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}}{s_1 + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}} + \frac{K_{droop,i}}{T_R K_{inertia,i}} \frac{1}{s_1 + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}}) \\
& = M \cdot (1 + \frac{\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}}{-\zeta \omega_n + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}} + j \omega_n \sqrt{1-\zeta^2}} + \frac{K_{droop,i}}{T_R K_{inertia,i}} \frac{\frac{1}{\omega_n}(-\zeta - j \sqrt{1-\zeta^2})}{-\zeta \omega_n + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}} + j \omega_n \sqrt{1-\zeta^2}}) \\
& = M \cdot (1 + \frac{(\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}})(-\zeta \omega_n + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}} - j \omega_n \sqrt{1-\zeta^2})}{(-\zeta \omega_n + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}})^2 + \omega_n^2(1-\zeta^2)} \\
& \quad + \frac{K_{droop,i}}{T_R K_{inertia,i}} \frac{\frac{1}{\omega_n}(-\zeta - j \sqrt{1-\zeta^2})(-\zeta \omega_n + \frac{k'_i}{2H_{w,i}} - j \omega_n \sqrt{1-\zeta^2})}{(-\zeta \omega_n + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}})^2 + \omega_n^2(1-\zeta^2)}) \\
& = M \cdot (1 + \frac{(\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}})(-\zeta \omega_n + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}} - j \omega_n \sqrt{1-\zeta^2})}{(-\zeta \omega_n + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}})^2 + \omega_n^2(1-\zeta^2)} \\
& \quad + \frac{K_{droop,i}}{T_R K_{inertia,i}} \frac{\frac{1}{\omega_n}(-\zeta(-\zeta \omega_n + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}}) - \omega_n(1-\zeta^2) - j \omega_n \zeta \sqrt{1-\zeta^2} - j(-\zeta \omega_n + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}})\sqrt{1-\zeta^2})}{(-\zeta \omega_n + \frac{3k_{MPPT} \omega_{r,i,0}}{2H_{w,i}})^2 + \omega_n^2(1-\zeta^2)})
\end{aligned} \tag{8}$$

Then it can be derived that:

$$B_1 = M \cdot \left(-\frac{\left(\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}} \right)}{\left(-\zeta\omega_n + \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}} \right)^2 + \omega_n^2(1-\zeta^2)} + \frac{K_{droop,i}}{T_R K_{inertia,i}} \frac{\frac{1}{\omega_n^2}(-\omega_n\zeta - (-\zeta\omega_n + \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}}))}{(-\zeta\omega_n + \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}})^2 + \omega_n^2(1-\zeta^2)} \right) \quad (9)$$

$$C_1 = \frac{M}{\omega_n \sqrt{1-\zeta^2}} \cdot \left(1 + \frac{\left(\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}} \right)(-\zeta\omega_n + \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}})}{(-\zeta\omega_n + \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}})^2 + \omega_n^2(1-\zeta^2)} \right) + \frac{K_{droop,i}}{T_R K_{inertia,i}} \frac{\frac{1}{\omega_n}(-\zeta(-\zeta\omega_n + \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}}) - \omega_n(1-\zeta^2))}{(-\zeta\omega_n + \frac{3k_{MPPT}\omega_{r,i,0}}{2H_{w,i}})^2 + \omega_n^2(1-\zeta^2)} \quad (10)$$

APPENDIX II

This part provide the results of the numerical experiments of turbine's rotational speed response with different system damping ratio ζ and natural frequency ω_n , which is shown in Fig. 1.

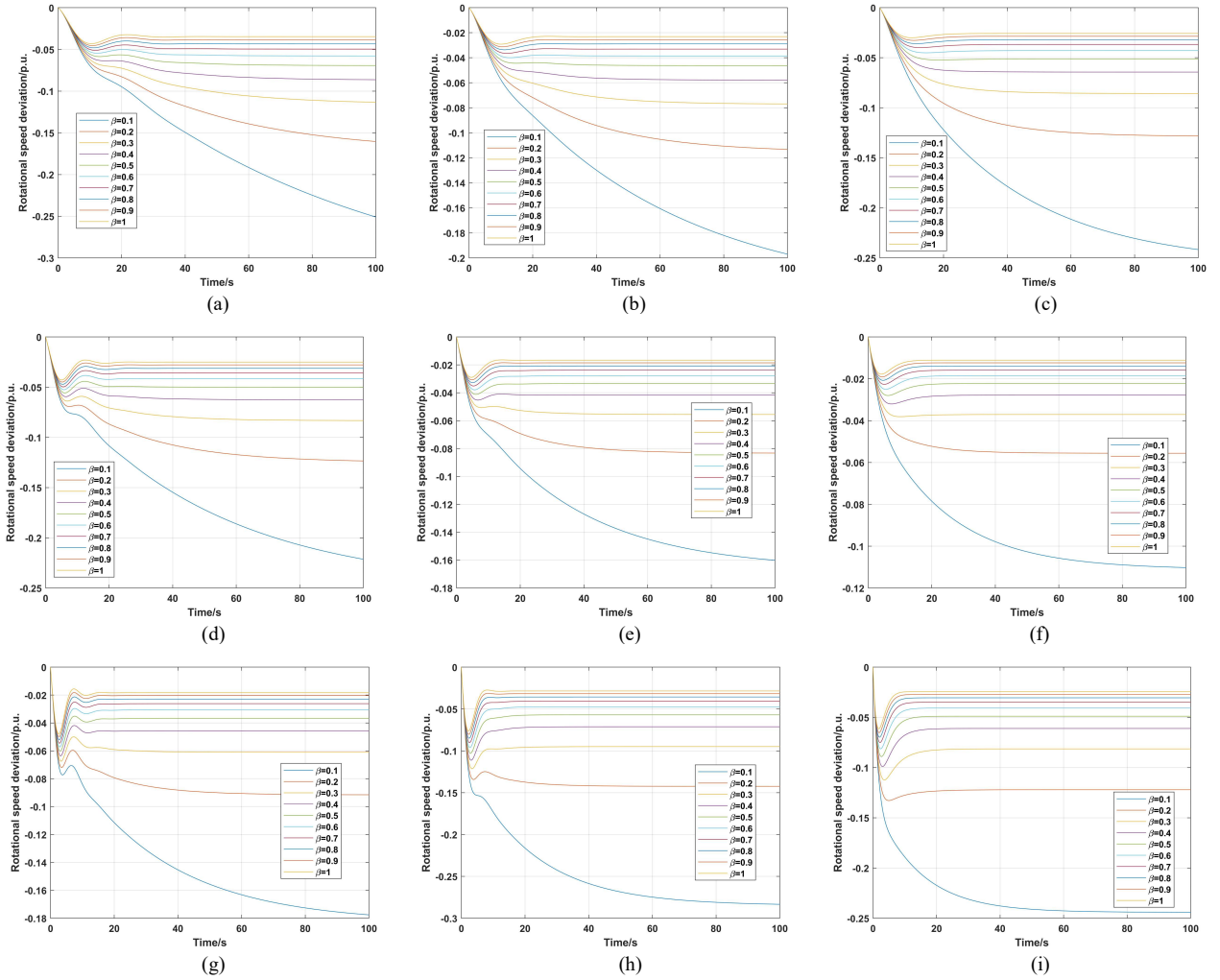


Fig. 1. Turbine's rotational speed dynamics with different ζ and ω_n

- (a). $\omega_n = 0.3, \zeta = 0.4$, (b). $\omega_n = 0.3, \zeta = 0.6$, (c). $\omega_n = 0.3, \zeta = 0.9$, (d). $\omega_n = 0.5, \zeta = 0.4$, (e). $\omega_n = 0.5, \zeta = 0.6$, (f). $\omega_n = 0.5, \zeta = 0.9$,
(g). $\omega_n = 0.8, \zeta = 0.4$, (h). $\omega_n = 0.8, \zeta = 0.6$, (i). $\omega_n = 0.8, \zeta = 0.9$

APPENDIX III

This part provides the maximum deviation of turbine's rotational speed $\Delta\omega_{r,i,\max}$ when $\beta = \infty$.

When a sudden increase in the load $\Delta P_L/s$ happens, $\Delta\omega_{r,i}(s)$ is obtained as follows:

$$\Delta\omega_{r,i}(s) = -\frac{\Delta P_L K_{inertia,i} R T_R \omega_n^2}{3k_{MPPT} \omega_{r,i,0}^2 (D_{eq} R + 1)} \cdot \frac{(s + \frac{K_{droop,i}}{K_{inertia,i}})(s + \frac{1}{T_R})}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad (11)$$

Denote $-\frac{\Delta P_L K_{inertia,i} R T_R \omega_n^2}{3k_{MPPT} \omega_{r,i,0}^2 (D_{eq} R + 1)} := L$. Then we can get:

$$\begin{aligned} \Delta\omega_{r,i}(s) &= L \cdot \frac{(s + \frac{K_{droop,i}}{K_{inertia,i}})(s + \frac{1}{T_R})}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \\ &= L \cdot \frac{1}{s} \left(\frac{(s + \frac{K_{droop,i}}{K_{inertia,i}})(s + \frac{1}{T_R})}{(s^2 + 2\zeta\omega_n s + \omega_n^2)} \right) \\ &= L \cdot \frac{1}{s} \left(\frac{s^2 + (\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R})s + \frac{K_{droop,i}}{T_R K_{inertia,i}}}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right) \\ &= L \cdot \frac{1}{s} \left(\frac{s^2 + 2\zeta\omega_n s + \omega_n^2 + (\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - 2\zeta\omega_n)s + \frac{K_{droop,i}}{T_R K_{inertia,i}} - \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right) \\ &= L \cdot \frac{1}{s} \left(1 + \frac{(\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - 2\zeta\omega_n)s + \frac{K_{droop,i}}{T_R K_{inertia,i}} - \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right) \\ &= \frac{L}{s} + \frac{L}{(\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - 2\zeta\omega_n)} \cdot \frac{s + \frac{\frac{K_{droop,i}}{T_R K_{inertia,i}} - \omega_n^2}{(\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - 2\zeta\omega_n)}}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \end{aligned} \quad (12)$$

It can be seen that, this response is similar to the system frequency response.

Denote $\frac{\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - 2\zeta\omega_n}{\frac{K_{droop,i}}{T_R K_{inertia,i}} - \omega_n^2} = T'_R$. Then the rotational speed nadir can be derived:

$$\Delta\omega_{r,i,\max} = L + \frac{L}{(\frac{K_{droop,i}}{K_{inertia,i}} + \frac{1}{T_R} - 2\zeta\omega_n)\omega_n^2} (1 + \sqrt{1 - \zeta'^2} \alpha' e^{-\zeta\omega_n t'_{nadir}}) \quad (13)$$

where,

$$\alpha' = \sqrt{\frac{1 - 2T'_R \zeta \omega_n + T_R'^2 \omega_n^2}{1 - \zeta'^2}} \quad (14)$$

$$t'_{nadir} = \frac{1}{\omega_d} \arctan\left(\frac{\omega_d T'_R}{\zeta\omega_n T'_R - 1}\right) \quad (15)$$

$$\omega_d = \omega_n \sqrt{1 - \zeta'^2} \quad (16)$$

APPENDIX IV

This part verifies the effectiveness of using $T_{f,i} = \frac{1}{\zeta\omega_n}$ as the coefficient for the low-pass filter.

Assume the $T_{f,i}$ is set as follows:

$$T_{f,i} = \frac{K}{\zeta\omega_n} \quad (17)$$

where K represents the ratio of the real pole of the transfer function to the real part of the conjugate complex poles.

When ζ is relatively small ($\zeta < 0.5$), it mainly means that the power system has insufficient rotational inertia. At this time, the system will have a more violent frequency response to changes in active power, as shown in Fig. 2. Different curves represent different ratio K . Here $\zeta = 0.25$.

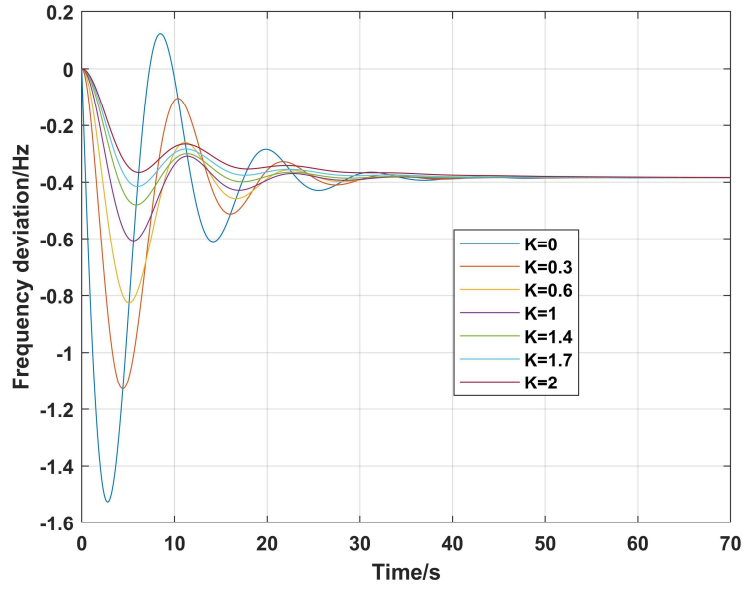


Fig. 2. System frequency response with different K when $\zeta = 0.25$

The maximum frequency deviation and the settling time are shown in Table 1. Here the settling time is the time it takes for the error between f and its steady-state value f_s to stay below 5% of $|f_s|$.

TABLE 1
THE MAXIMUM FREQUENCY DEVIATION AND SETTLING TIME WITH DIFFERENT K WHEN $\zeta = 0.25$

Index	$K = 0$	$K = 0.3$	$K = 0.6$	$K = 1$	$K = 1.4$	$K = 1.7$	$K = 2$
Maximum frequency deviation/Hz	-1.5269	-1.1278	-0.8261	-0.6072	-0.4802	-0.4152	-0.3835
Settling time/s	21.7187	23.3502	23.6280	19.2972	22.9968	25.2791	27.9286

It can be seen from Fig. 2 and Table 1 that, in this situation (ζ is relatively small), selecting $K = 1$ appears to be a judicious choice. The result indicates that when $K = 1$, the system achieves the minimum settling time. Although the maximum frequency deviation at $K = 1$ is not the smallest, it is significantly less than that at $K = 0$, suggesting a considerable improvement in the system's ability to handle frequency overshoot. Therefore, $K = 1$ emerges as an optimal trade-off between minimizing frequency overshoot and ensuring a swift return to equilibrium.

When ζ is relatively large ($\zeta > 0.5$), it indicates that the power system has sufficient rotational inertia. At this time, the system will have a less violent frequency response to changes in active power, as shown in Fig. 3. Different curves represent different ratio K . Here $\zeta = 0.75$.

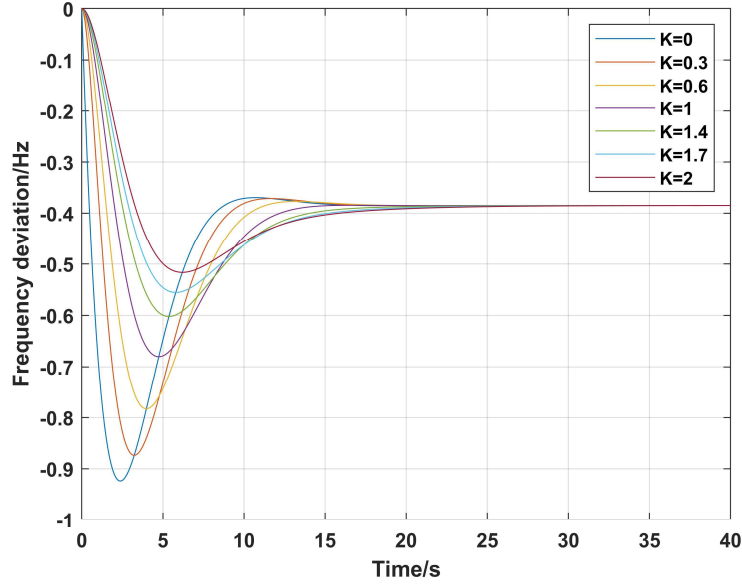


Fig. 3. System frequency response with different K when $\zeta = 0.75$

The maximum frequency deviation and the settling time are shown in Table 2.

TABLE 2
THE MAXIMUM FREQUENCY DEVIATION AND SETTLING TIME WITH DIFFERENT K WHEN $\zeta = 0.75$

Index	$K = 0$	$K = 0.3$	$K = 0.6$	$K = 1$	$K = 1.4$	$K = 1.7$	$K = 2$
Maximum frequency deviation/Hz	-0.9242	-0.8739	-0.7828	-0.6801	-0.6020	-0.5554	-0.5165
Settling time/s	7.8563	8.7989	10.1360	11.9003	13.4937	14.3935	14.8604

We can see that in this situation (ζ is relatively large), $K = 1$ also achieves a balance between maximum frequency deviation and settling time. At $K = 1$, the system demonstrates a favorable compromise: the maximum frequency deviation is significantly reduced compared to lower K values, indicating a robust resistance to frequency fluctuation. Simultaneously, the settling time remains within a reasonable range, preventing the excessive delays in system frequency response that are evident at higher K values.

In summary, whether ζ is large or small, setting $K = 1$ consistently appears to be a wise choice. This particular setting strikes a useful balance regardless of the system's inherent level of damping, as indicated by the value of ζ . For a low ζ , $K = 1$ reduces the system's frequency overshoot to a manageable level while also achieving a fast settling time. In a system with high ζ , the $K = 1$ continues to provide a stable balance between small frequency deviation and a reasonable settling time. This demonstrates that $K = 1$ can adapt to power system with different characteristics, offering a middle ground that aligns with both a quick response to power change in load and less frequency overshoot. Consequently, $K = 1$ emerges as an effective choice across varying conditions of system inertia.

APPENDIX V

This part shows the reason to select the $K(\zeta_i)$ in the estimation formula of $\Delta T_{rec,i}$ as follows:

$$K(\zeta_i) = \begin{cases} 3 & \zeta_i < 0.5 \\ 8\zeta_i - 1 & \zeta_i \geq 0.5 \end{cases} \quad (18)$$

Here $\Delta T_{rec,i}$ is defined as the time it takes for the error between f and its steady-state value f_s to stay below 5% of $|f_s|$. Fig. 4 shows the relationship of $K(\zeta_i)$ as a function of ζ_i when $\omega_n = 0.57$, $T_R = 6.5s$.

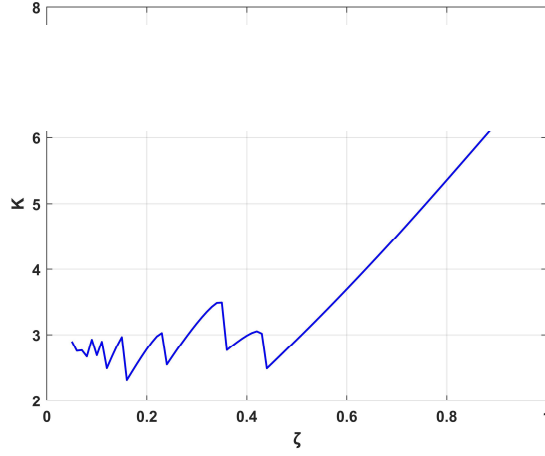


Fig. 4. $K(\zeta_i)$ with respect to ζ_i when $\omega_n = 0.57$, $T_R = 6.5s$

It can be observed that for $\zeta_i < 0.5$, $K(\zeta_i)$ fluctuates around the value of 3. For $\zeta_i \geq 0.5$, there is a clear linear increase in $K(\zeta_i)$ with respect to ζ_i and this relationship can be determined by $K(\zeta_i) \approx 8\zeta_i - 1$.

Then we plot the figure of $K(\zeta_i)$ as a function of ζ_i for different ω_n and T_R in Fig. 5 and Fig. 6.

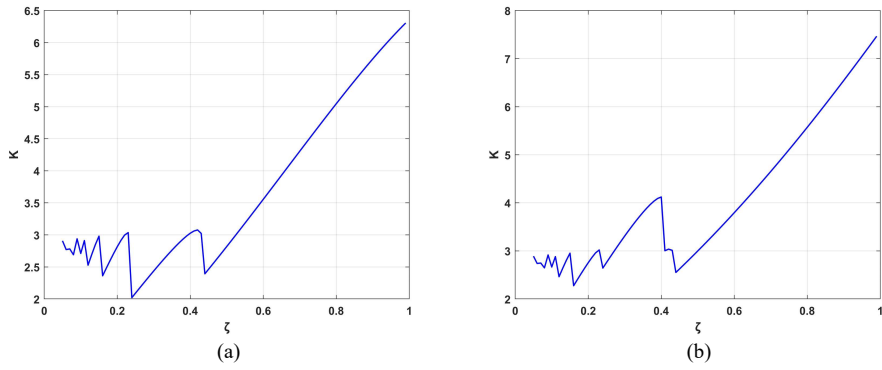


Fig. 5. $K(\zeta_i)$ with respect to ζ_i for different ω_n , (a). $\omega_n = 0.43, T_R = 6.5s$, (b.) $\omega_n = 0.72, T_R = 6.5s$

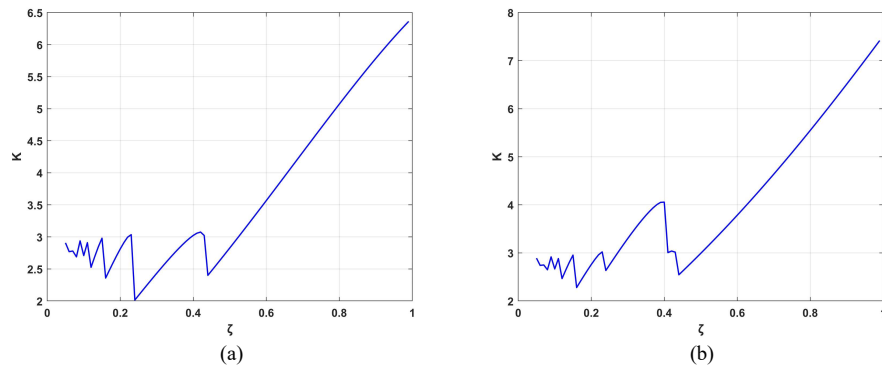


Fig. 6. $K(\zeta_i)$ with respect to ζ_i for different ω_n , (a). $\omega_n = 0.57, T_R = 5s$, (b.) $\omega_n = 0.57, T_R = 8s$

From Fig. 5 and Fig. 6, it's evident that while discrepancies do exist, they are minimal and do not significantly detract from the overall validity of our estimation formula. The formula consistently captures the general trend of $K(\zeta_i)$ as a function of ζ_i . Therefore, (18) remains a useful tool for estimating the parameter $K(\zeta_i)$.